

SAMUEL LIHN

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EDUCATION

The Johns Hopkins University

August 2023 – May 2027

B.S. in Mechanical Engineering; M.S. in Robotics – Controls and Dynamical Systems Track

Baltimore, MD

Relevant Coursework: Learning-Based Control for Robotics, Optimal Control for Space Systems, Robot Kinematics and Dynamics

EXPERIENCE

Zoox

May 2026 – August 2026

Robot Dynamics Simulation Engineer Intern

Foster City, CA

- Developed vehicle-model fidelity metrics using magnitude and phase errors and statistical distribution comparisons to diagnose braking discrepancies; validated metric sensitivity through controlled model-parameter perturbations.
- Designed and deployed a Python vehicle-model validation pipeline that automatically compared simulation against track-test telemetry on every pull request, enabling regression detection between milestone validation sprints.
- Built and deployed a Streamlit dashboard adopted by vehicle dynamics and firmware validation teams to assess model changes and support PR review, with automated reporting and on-demand validation.

Autonomous Control and Exploration Lab – Johns Hopkins LCSR

August 2025 – Present

Undergraduate Researcher

Baltimore, MD

- Built a free-floating satellite manipulation environment in MuJoCo/MJX and adapted Hydrax sampling-based MPC to plan coordinated base-arm motion for autonomous grasping.
- Trained flow-matching policies from MPPI trajectories using Generative Predictive Control; integrated policy warm starts with predictive sampling, reducing planning latency approximately 4× in preliminary simulation evaluations.
- Designed approach and grasping objectives and adapted sphere-based collision penalties to discourage contact with environmental obstacles.

Zoox

May 2025 – August 2025

Suspension Bench Test & Data Analysis Intern

Foster City, CA

- Analyzed three months of production-fleet telemetry, identifying 36% of active-suspension valve cycles as potential reduction opportunities associated with negligible actuator travel or stationary operation.
- Linked intermittent valve jitter to command-signal quantization; evaluated filtering and recommended supplier-supported smooth control and vehicle-state-based suspension gating to reduce unnecessary cycling.
- Implemented a simulated suspension controller in ECU-TEST and executed HIL tests on a rig containing physical suspension hardware and controllers.

Fulfil Solutions

June 2024 – August 2024

Systems Engineering Intern

Plymouth Meeting, PA

- Commissioned grocery induction and storage gantry systems for an initial retail deployment, integrating tray scanning, transfer, and robotic placement into storage racks.
- Calibrated gantry alignment, flashed motor controllers, and diagnosed embedded Linux networking issues; resolved mechanical and electrical integration problems to advance deployment readiness.

Baja SAE at Johns Hopkins

August 2023 – July 2025

Controls & DAQ Systems Integration Engineer

Baltimore, MD

- Integrated vehicle DAQ electronics and wiring harnesses within tight mass and packaging constraints, addressing water and mud ingress, abrasion, and vibration; validated through shop ingress testing and vehicle test-day operation.
- Designed passive cooling for an enclosed Raspberry Pi using a custom heat spreader and heat pipe coupled to an external heat sink, enabling heat rejection without enclosure ventilation.
- Designed a lightweight rack-and-pinion steering assembly that was manufactured and raced; addressed housing loads through structural analysis and reduced sensing-system mass by replacing an encoder with a Hall-effect sensor.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Controls & Optimization: MPC, MPPI, trajectory optimization, constrained optimization, quadratic programming, CasADi, CVXPY

Dynamics & Robotics: Rigid-body dynamics, MuJoCo/MJX, Isaac Sim, Simulink, ROS, HIL testing

Machine Learning: JAX (vectorization), PyTorch, flow matching

Software & Hardware: Linux, Git, Docker, CI/CD, CAN, SPI/I2C

Data & Analysis: NumPy, SciPy, pandas, Databricks, Apache Spark, signal processing